KW 10: Financial Markets

Ch 4, Ch 5.2, Ch 5.3

Blanchard - Macroeconomics, 7th Global Edition

Study Notes

CHAPTER 4: FINANCIAL MARKETS I

Financial markets determine the cost of funds for firms, households, and the government. The chapter simplifies to two financial assets: money (pays no interest) and bonds (pay interest rate i). Focus is on how the interest rate is determined and the role of the central bank.

Semantic Traps: Money, Income, and Wealth

Money: Currency + checkable deposits. Used for transactions. A STOCK variable.

Income: What you earn (wages + interest + dividends). A FLOW variable.

Saving: Part of after-tax income not spent. A FLOW.

Financial wealth: All financial assets minus liabilities. A STOCK.

Investment: Purchase of NEW capital goods (NOT financial assets).

Financial investment: Purchase of shares/bonds.

4-1 The Demand for Money

The Portfolio Choice: Money vs. Bonds

- **Money**: Can be used for transactions, pays no interest. Two types: currency (coins/bills) and checkable deposits (bank accounts with checks/debit cards)
- **Bonds**: Pay positive interest rate i, but cannot be used directly for transactions. Transaction costs exist for buying/selling bonds.
- You should hold BOTH: enough money for transactions, rest in bonds for interest

Two Key Determinants

- **(1) Level of transactions**: More transactions -> need more money on hand. Roughly proportional to nominal income ($Y).
- **(2) Interest rate on bonds**: Higher i -> more willing to deal with inconvenience of selling bonds -> hold less money. At very high rates, minimize money holdings.

The Money Demand Equation

$$M^d = \$Y \cdot \underset{(-)}{L(i)}$$

$$M^d = \text{demand for money}$$

$$\$Y = \text{nominal income}$$

$$L(i) = \text{decreasing function of interest rate}$$

- Money demand increases **proportionally** to nominal income

- Money demand depends **negatively** on the interest rate
- What matters is NOMINAL income (not real) - if prices double, need twice as much money for same transactions

Figure 4-1: Money Demand Curve

- Downward sloping: lower interest rate -> more money demanded
- An increase in nominal income shifts the entire curve RIGHT (more money demanded at every interest rate)

Who Holds U.S. Currency?
Total U.S. currency in circulation: $750 billion (2006)
U.S. households held only $170 billion
~$500 billion (66%) held by FOREIGNERS
Countries like Russia ($80B), Argentina ($50B+) hold dollars as safe assets
Result: The rest of the world makes an interest-free loan to the U.S.

4-2 Determining the Interest Rate: I

Assumes only money is currency (central bank money). Checkable deposits introduced in 4-3.

Equilibrium Condition

$$M = \$Y \cdot L(i)$$

- The interest rate i must be such that people are willing to hold exactly the existing money supply M

Figure 4-2: Interest Rate Determination

- Money demand curve: downward-sloping
- Money supply curve: vertical line at M
- Equilibrium at intersection point A

Effects of Changes in Nominal Income (Figure 4-3)

- Increase in nominal income -> money demand shifts RIGHT -> interest rate INCREASES
- Reason: at old rate, money demand > supply. Higher i reduces demand back to equilibrium

Effects of Increase in Money Supply (Figure 4-4)

- Increase in M -> money supply shifts RIGHT -> interest rate DECREASES
- Reason: lower i increases demand to match the now larger supply

Monetary Policy and Open Market Operations

- Central bank changes money supply by **buying or selling bonds in the bond market**
- **Open market operations**: transactions in the 'open market' for bonds

Open Market Operations
EXPANSIONARY OMO: Central bank BUYS bonds, pays with newly created money
-> Money supply increases -> Bond prices increase -> Interest rate DECREASES

CONTRACTIONARY OMO: Central bank SELLS bonds, removes money from circulation
-> Money supply decreases -> Bond prices decrease -> Interest rate INCREASES

Central Bank Balance Sheet (Figure 4-5)

- Assets: Bonds held by the central bank
- Liabilities: Money (currency) in the economy
- OMOs lead to equal changes in assets and liabilities

Bond Prices and Bond Yields

Consider a one-year bond (T-bill) with face value $100:

$$i = \frac{\$100 - \$P_B}{\$P_B}$$

$$\$P_B = \frac{\$100}{1+i}$$

- Higher bond price -> lower interest rate (and vice versa)
- 'Bond markets went up' = bond prices rose = interest rates fell

Choosing Money or Choosing the Interest Rate?

- The two descriptions are equivalent:
  > (1) Central bank chooses money supply, interest rate adjusts to equilibrium

  > (2) Central bank chooses interest rate, adjusts money supply to achieve it
- **Modern central banks think in terms of choosing the interest rate**
- News says: 'The Fed decided to increase the interest rate' (not 'decrease the money supply')

4-3 Determining the Interest Rate: II (With Banks)

In reality, money includes currency AND checkable deposits. Checkable deposits are supplied by private banks, not the central bank.

What Banks Do

- Banks are **financial intermediaries** - receive funds and use them to buy assets or make loans
- Special because their liabilities are money (checkable deposits)

Bank Balance Sheet (Figure 4-6)

Assets	Liabilities
Reserves	Checkable deposits
Loans (~70%)	
Bonds (~30%)	

Reserves

- Banks keep some funds as reserves (cash + deposits at central bank)
- Three reasons: (1) daily cash flow mismatches, (2) interbank check clearing, (3) **reserve requirements**

- U.S. reserve requirement: at least **10% of checkable deposits**

Central Bank Money (The Monetary Base)

$$H = CU + R \quad \text{(Central Bank Money = Currency + Reserves)}$$

Demand for Central Bank Money

Simplifying assumption: people hold only checkable deposits (no currency).

- Demand for checkable deposits = demand for money by people:

$$M^d = \$Y \cdot L(i)$$

Demand for reserves by banks (theta = reserve ratio):

$$H^d = \theta \cdot M^d = \theta \cdot \$Y \cdot L(i)$$

Equilibrium

$$H = \theta \cdot \$Y \cdot L(i)$$

- H = supply of central bank money (controlled by central bank via OMOs)
- Same qualitative results as before: increase in H -> lower interest rate

Federal Funds Market and Federal Funds Rate

In the U.S., banks trade reserves in the FEDERAL FUNDS MARKET.
The interest rate in this market = FEDERAL FUNDS RATE.
This is the main indicator of U.S. monetary policy.
Changes in the federal funds rate make front-page news.

4-4 The Liquidity Trap

The Zero Lower Bound

- The interest rate on bonds **cannot be negative** -> the **zero lower bound (ZLB)**
- When i = 0, monetary policy cannot decrease it further -> **liquidity trap**

Why the Zero Lower Bound Exists

- At i = 0, both money and bonds pay the same interest rate: zero
- People are **indifferent** between holding money or bonds
- Additional money supply is simply absorbed without lowering rates

Figure 4-8: The Liquidity Trap

- Money demand curve becomes **horizontal at i = 0**

- If money supply is at/beyond the flat portion: interest rate = 0
- Further increases in money supply have NO effect on interest rate
- Monetary policy is **ineffective** in the liquidity trap

The Liquidity Trap in Practice
During the 2008 crisis:
- Bank of England cut rate from 5% to 0.5%
- Bank reserves expanded from 23.6B to 265.5B pounds
- Rate remained at 0.5% despite massive money expansion
- This is exactly what liquidity trap theory predicts

Appendix: Both Currency and Checkable Deposits

When people hold proportion c in currency and (1-c) in checkable deposits:

$$CU^d = c \cdot M^d, \quad D^d = (1-c) \cdot M^d, \quad R^d = \theta(1-c) \cdot M^d$$

Demand for central bank money:

$$H^d = [c + \theta(1-c)] \cdot \$Y \cdot L(i)$$

Equilibrium:

$$H = [c + \theta(1-c)] \cdot \$Y \cdot L(i)$$

The Money Multiplier

$$M = \frac{H}{c + \theta(1-c)} = \text{Money Multiplier} \times H$$

- Money multiplier is always > 1:

$$\text{Money multiplier} = \frac{1}{c + \theta(1-c)} > 1$$

- Example:

$$c = 0.4, \ \ \theta = 0.1 \quad \Longrightarrow \quad \text{multiplier} = \frac{1}{0.46} = 2.2$$

- Each dollar of central bank money leads to $2.20 of total money

CHAPTER 5, SECTION 5-2: Financial Markets and the LM Relation

The LM Relation

From Chapter 4, the interest rate is determined by money market equilibrium:

$$M = \$Y \cdot L(i)$$

Dividing both sides by the price level P to get real terms:

$$\frac{M}{P} = Y \cdot L(i)$$

- **Real money supply** = money stock in terms of goods:

$$\text{Real money supply} = \frac{M}{P}$$

- **Real money demand** depends on real income Y and interest rate i

Deriving the LM Curve

Two approaches:

Traditional Approach (choosing M)

- Central bank chooses M, price level P is fixed in short run -> M/P is given
- If real income increases -> money demand increases -> interest rate must rise
- Result: upward-sloping LM curve (traditional)

Modern Approach (choosing i)

- Modern central banks directly choose the interest rate i-bar
- They adjust M to achieve whatever money demand arises at that rate
- Result: **LM curve is a horizontal line at i-bar**

Figure 5-4: The LM Curve

The LM curve is a HORIZONTAL LINE at the interest rate i-bar
chosen by the central bank.
The central bank adjusts money supply to maintain i-bar
regardless of the level of output.

CHAPTER 5, SECTION 5-3: Putting the IS and LM Relations Together

The IS-LM Model

Two relations that jointly determine output and the interest rate:

$$\text{IS:} \quad Y = C(Y - T) + I(Y, i) + G$$

$$\text{LM:} \quad i = \bar{i}$$

Figure 5-5: The IS-LM Model

- IS curve: downward sloping (higher i -> lower Y)
- LM curve: horizontal line at i-bar
- Equilibrium at intersection (point A): both goods market and financial markets are in equilibrium

Fiscal Policy in the IS-LM Model

An Increase in Taxes (Figure 5-6)

- Higher T -> lower disposable income -> lower consumption -> lower demand -> lower output
- IS curve shifts LEFT (from IS to IS')
- LM curve unchanged (central bank keeps i-bar)
- Result: Output DECREASES, interest rate UNCHANGED
- Similarly: increase in G shifts IS RIGHT -> output increases, rate unchanged

Monetary Policy in the IS-LM Model

An Increase in the Interest Rate (Figure 5-7)

- Central bank raises rate from i-bar to i-bar'
- IS curve does NOT shift
- LM curve shifts UP (from LM to LM')
- Economy moves along the IS curve to new equilibrium
- Result: Output DECREASES, interest rate INCREASES
- Mechanism: higher i -> lower investment -> lower demand -> lower output (multiplier effect)

Policy Mix: Summary of Effects

Policy Change	IS	LM	i	Y
Increase T	LEFT	Same	Same	DOWN

Decrease T	RIGHT	Same	Same	UP
Increase G	RIGHT	Same	Same	UP
Decrease G	LEFT	Same	Same	DOWN
Raise i-bar (tight)	Same	UP	UP	DOWN
Lower i-bar (loose)	Same	DOWN	DOWN	UP

Both fiscal policy (G, T shifting the IS curve) and monetary policy (i-bar shifting the LM curve) can affect output. The combination of these two policies - the policy mix - determines the overall outcome.

SUMMARY OF KEY EQUATIONS AND TERMS

$$(4.1) \quad M^d = \$Y \cdot L(i) \quad \text{Money demand}$$

$$(4.2) \quad M = \$Y \cdot L(i) \quad \text{Money market equilibrium}$$

$$(4.4) \quad H^d = \theta \cdot \$Y \cdot L(i) \quad \text{Reserves demand (no currency)}$$

$$(4.6) \quad H = \theta \cdot \$Y \cdot L(i) \quad \text{Central bank money equilibrium}$$

$$i = \frac{100 - P_B}{P_B} \quad \text{Interest rate from bond price}$$

$$P_B = \frac{100}{1+i} \quad \text{Bond price from interest rate}$$

$$(4.A9) \quad H = [c + \theta(1-c)] \cdot \$Y \cdot L(i) \quad \text{General equilibrium}$$

$$M = \frac{H}{c + \theta(1-c)} \quad \text{Money multiplier relation}$$

$$(5.2) \quad Y = C(Y-T) + I(Y, i) + G \quad \text{IS relation}$$

$$(5.3) \quad \frac{M}{P} = Y \cdot L(i) \quad \text{LM relation (real terms)}$$

$$(5.4) \quad i = \bar{i} \quad \text{LM curve (modern)}$$

Key Terms

- **Federal Reserve (Fed)**: The U.S. central bank
- **Open Market Operation (OMO)**: Central bank buying/selling bonds to change money supply
- **Expansionary OMO**: Buy bonds -> increase M -> decrease i

- **Contractionary OMO**: Sell bonds -> decrease M -> increase i
- **Treasury bill (T-bill)**: Government bond with maturity < 1 year
- **Central bank money**: Currency + Reserves (= Monetary Base = High-powered money)
- **Reserve ratio (theta)**: Reserves / Checkable deposits (min 10% in U.S.)
- **Federal funds rate**: Interest rate in the federal funds market; main U.S. policy indicator
- **Zero lower bound (ZLB)**: Interest rates cannot go below zero
- **Liquidity trap**: At i=0, more money has no effect on interest rate
- **IS curve**: Downward-sloping: all (i,Y) pairs where goods market is in equilibrium
- **LM curve**: Horizontal line at i-bar (modern); financial market equilibrium
- **Money multiplier**: total money as multiple of central bank money
- **Real money supply**: money stock measured in terms of goods
- **Policy mix**: Combination of fiscal and monetary policy

Key formulas for reference:

$$\text{Money multiplier} = \frac{1}{c + \theta(1 - c)}$$

$$\text{Real money supply} = \frac{M}{P}$$